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Alan P. Boyle

Education

2005–2009 **Doctor of Philosophy,** Computational Biology and Bioinformatics
Duke University, Durham, NC

2001–2005 **Bachelor of Science,** *summa cum laude,* Biochemistry and Molecular Biology
Bachelor of Science, *summa cum laude,* Computer Science
Mississippi State University, Starkville, MS

Academic Appointments

2025–present **Professor with tenure,** Department of Computational Medicine & Bioinformatics
Professor, Department of Human Genetics
University of Michigan, Ann Arbor, MI

2020–2025 **Associate Professor with tenure,** Department of Computational Medicine & Bioinformatics
Associate Professor, Department of Human Genetics
University of Michigan, Ann Arbor, MI

2015–2020 **Assistant Professor,** Department of Human Genetics
2014–2020 **Assistant Professor,** Department of Computational Medicine & Bioinformatics
University of Michigan, Ann Arbor, MI

2010–2014 **Postdoctoral Scholar,** Genetics
Stanford University, Stanford, CA; Advisor: Dr. Michael Snyder

Spring 2010 **Postdoctoral Associate,** Computational Biology
Duke University, Durham, NC; Advisor: Dr. Terrence S. Furey

Institutional Affiliations

2023–present **Core Member,** Rogel Cancer Center
2021–present **Affiliate,** Michigan Neuroscience Institute
2016–present **Member,** Center for RNA Biomedicine
2020–2023 **Affiliate Member,** Rogel Cancer Center

Other Professional Experience

2013–2014 **Consultant,** Color Genomics
Personalized medicine and genomics startup

Scholarships, Fellowships, and Honors

2022 Valuing our Own Award, Michigan Medicine
2019 Endowment for the Basic Sciences Teaching Award
2018 First Place in CAGI5 Regulation Saturation Challenge
2017 NSF CAREER Award
2016 Institutional nominee for W.M. Keck Foundation Medical Science Research Program
2016 Institutional nominee for Searle Scholar Award
2015–2017 Alfred P. Sloan Foundation Fellowship in Computational & Evolutionary Molecular Biology
2013–2014 NIH Pathway to Independence Award (K99/R00) [1K99HG007356-01]
2012 AAAS/Science Program for Excellence in Science
2005–2008 NSF Graduate Research Fellowship
2005–2009 James B. Duke Fellowship

Summer 2004	Mayo Clinic Summer Undergraduate Research Fellow
2003	Barry M. Goldwater Memorial Scholarship
Summer 2003	The Institute for Genomic Research (TIGR) Summer Fellow
2001	Robert C. Byrd Honors Scholarship
2001	Mississippi State University Presidential Scholarship
2001	National Merit Scholarship

Research Support

Active: Principal Investigator and Multi-Principal Investigator

2026–2031 R01 NS145291, NIH/NINDS (Multi-PI: Mills, Boyle)
Characterization and functional impact of somatic numtogenesis in the human cortex
Characterizes somatic nuclear mitochondrial DNA insertions in the human cortex and tests their effects on gene expression, chromatin structure, aging, and brain function.

2026–2027 National Ataxia Foundation Research Grant (Co-PI: Burmeister, Boyle)
FGF14 intronic GAA repeats in vestibular problems beyond SCA27B
Examines genetic features of the FGF14 repeat that may underlie subclinical or atypical expansions.

2023–2028 UG3/UH3 NS132084, NIH/Office of the Director (Multi-PI: Mills, Boyle)
Molecular and Computational Tools for Identifying Somatic Mosaicism in Human Tissues
Develops long-read molecular and computational methods to identify somatic mosaicism in normal human tissues as part of the SMaHT Consortium.

2022–2026 R01 GM144484, NIH/NIGMS (PI: Boyle)
Mobile element derived chromatin looping variability in human populations
Defines how polymorphic LTR13 integrations alter three-dimensional chromatin conformation and gene regulation.

2021–2027 U01 HG011952, NIH/NHGRI (PI: Boyle)
Predicting the impact of genomic variation on cellular states
Develops models and tools that use single-cell data to predict the effects of genomic variation as part of the IGVF Consortium.

Active: Co-Investigator and Consultant

2024–2029 R01 DE032699, NIH/NIDCR (Multi-PI: Brenner, Mills, Spector; Co-I with effort)
Defining the Role of HPV Integration Structures in HNSCC Molecular Heterogeneity

2023–2028 R01 NS099280, NIH/NINDS (PI: Todd; Consultant)
Hexanucleotide repeat translation in ALS and Frontotemporal Dementia

2022–2027 R01 CA260677, NIH/NCI (PI: Malek; Co-I with effort)
The Biology of Mutant STAT6 in Follicular Lymphoma

2022–2026 R01 NS122165, NIH/NINDS (PI: Castro; Co-I with effort)
Uncover the role of H3.3-G343R mutation in shaping the DNA damage response, anti-tumor immunity, and mechanisms of resistance in glioma

Completed: Principal Investigator and Multi-Principal Investigator

2022–2025 R21 CA257864, NIH/NCI (PI: Boyle)
High-throughput inverted reporter assay for characterization of silencers and enhancer blockers

2022–2025 Taubman Institute Innovation Program, University of Michigan (Co-PI: Todd, Boyle, Mills)
Repetitive elements in human health and disease

2017–2025 U41/U24 HG009293, NIH/NHGRI (Multi-PI: Boyle, Cherry)
RegulomeDB: A Resource for the Human Regulome

2020–2023 R21 HG011493, NIH/NHGRI (Multi-PI: Boyle, Mills)
New technologies for accurate capture and sequencing of repeat-associated regions

2021–2023 Cancer Center Discovery Award, University of Michigan (PI: Boyle)
Direct capture of complete HPV integration sites using long-read sequencing

2022 R21 HG011493-02S1, NIH/NIA (Multi-PI: Boyle, Mills)
New technologies for accurate capture and sequencing of repeat-associated regions (supplement)

2022 NVIDIA GPU Grant, NVIDIA Corporation (PI: Boyle)

2017–2022 DBI-1651614, NSF/BIO/DBI (PI: Boyle)
CAREER: Conservation of cohesin-containing cis-regulatory modules in the human and mouse lineages

2019–2020 Precision Health Investigators Award, University of Michigan (Co-PI: Todd, Boyle, Mills)
Short Tandem Repeats in Precision Health and Human Disease

2019 NVIDIA GPU Grant, NVIDIA Corporation (PI: Boyle)

2017–2018 Eleanor and Larry Jackier U-M/Technion and Weizmann Collaborative Research Grant (Co-PI: Boyle, Mandel-Gutfreund)
Michigan-Israel Partnership for Research and Education
Identifying novel disease-related mutations in the genomic environments around transcription factor binding sites

2015–2017 FG-2015-65465, Alfred P. Sloan Foundation (PI: Boyle)
Sloan Research Fellowship in Computational and Evolutionary Molecular Biology

2013–2017 R00 HG007356 Pathway to Independence Award (K99/R00), NIH/NHGRI (PI: Boyle)
Global Discovery and Validation of Functional Regulatory Elements

Completed: Co-Investigator and Consultant

2018–2026 R01 HD093570, NIH/NICHD (PI: Bielas; Co-I with effort)
Genetic Diagnosis of Neurodevelopmental Disorders in India

2021–2026 K08 HL153799, NIH/NHLBI (PI: Denstaedt; Consultant)
Predisposition for Lung Injury in Sepsis Survival

2021–2025 R01 HD104680, NIH/NICHD (PI: Hammoud; Co-I with effort)
Sperm Chromatin: Implications on organismal development and fertility

2023–2025 Michigan Alzheimer's Disease Center Developmental Project, University of Michigan (PI: Zhou; Consultant)
Explore the functional impact of transposable elements in Alzheimer's disease and related dementias

2017–2024 R35 HL135824, NIH/NHLBI (PI: Willer; Co-I with effort)
Using Genetics to Inform Mechanism of Cardiovascular Disease

2020–2024 W81XWH2010336, U.S. Department of Defense/Army (PI: Aguilar; Co-I with effort)
Understanding and Enhancing the Regenerative Capacity of Skeletal Muscle to Trauma by Targeting Muscle-Nerve Synergy

2016–2020 R01 HL130705, NIH/NHLBI (PI: Willer; Co-I with effort)
Large-scale human genetics to understand molecular mechanisms of atrial fibrillation and related traits

Professional Service

Institutional and Consortium Leadership

2026–present Michigan Neuroscience Institute Executive Committee

2022–present	University of Michigan Biomedical Research Council [Standing Member; Co-Chair 2026–present]
2014–2018, 2025–present	DCMB Admissions Committee [Chair 2025–present]
2024–present	Department of Human Genetics Faculty Development Committee
2023–present	R01 Bootcamp Medical School Cohort Coach
2023–present	Somatic Mosaicism across Human Tissues (SMaHT) Consortium Steering Committee
2021–present	Impact of Genomic Variation on Function (IGVF) Consortium Steering Committee
2018–present	DCMB Laboratory Safety Liaison
2015, 2016, 2025, 2026	DCMB Retreat Planning Committee Chair, including the first annual retreat
2020–2022	Department of Human Genetics M.S. Admissions Committee
2017–2024	DCMB Preliminary Exam Abstract Review Committee [Chair 2018–2022]
2019–2020	Department of Human Genetics Ph.D. Admissions Committee
2017–2020	Department of Human Genetics Faculty Recruitment and Promotions Committee
2016–2020	DCMB Seminar Series Committee [Chair]
2018–2019	Cellular and Molecular Biology Admissions Committee
2017–2019	Endowment for the Basic Sciences Faculty IT Committee
2016–2019	DCMB Faculty Recruitment Committee
2015–2017	Department of Human Genetics Computational Support Committee
2008–2009	Duke Computational Biology and Bioinformatics Student Committee

Editorial and Program Committee Service

2026	Academic Editor, PLOS ONE
2026	Program Committee, Proceedings, ISMB/ECCB
2023	Program Committee, Genome Sequence Analysis, ISMB/ECCB
2023	Program Committee, Biomedical Informatics, ISMB/ECCB
2018, 2020	Program Committee, Comparative and Functional Genomics, ISMB/ECCB
2018, 2019	Program Committee, Studies of Phenotypes and Clinical Applications, ISMB/ECCB
2019	Program Committee, General Computational Biology, ISMB/ECCB
2017	Program Committee, Regulatory Genomics Special Interest Group Meeting (RegGenSIG), ISMB/ECCB
2015–2018	Program Committee, Great Lakes Bioinformatics and Canadian Computational Biology Conference (GLBIO/CCBC)
2015–2016	Program Committee, Algorithms for Computational Biology (ALCOB)
2013–2016	Program Committee, Gene Regulation and Transcriptomics, ISMB/ECCB
2012–2015	DNA Day Essay Contest Detailed Review Judge, American Society of Human Genetics

Grant Review

2026	NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)
2026	NIH Study Section ZRG1 IIDB Z (70) - Stage I New Innovator Award Program (mail review)
2026	NIH Study Section ZRG1 MGG-F (90) S - Special Topics in Genomics, Genetic Evolution, and Technology Development (ad hoc)
2025	NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)
2024	NIH Study Section ZRG1 BBBT-D (50) - PAR Panel: Biodata Management and Common Fund Data Sets (ad hoc; Co-Chair)
2024	NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)
2023	NSF Review Panel - Molecular and Cellular Biosciences, Genetic Mechanisms (ad hoc)
2023	NIH Study Section - Multi-Omics of Health and Disease Data Analysis and Coordination Center
2023	NIH Study Section GVE - Genetic Variation and Evolution Study Section (ad hoc)
2022	NIH Study Section ZRG1 ISB-S (57) - Academic-Industrial Partnerships for Translation of Technologies for Diagnosis and Treatment
2022	NASA Study Section E.11 Space Biology: Animal Studies - Omics Systems [21SBAS-OmicsSys] (ad hoc)
2020	NIH/NIMH Study Section ZMH1 ERB-C (08) - Fine-Mapping Genome-Wide Associated Loci to Identify Proximate Causal Mechanisms of Serious Mental Illness
2019	NIH/NIMH Study Section ZMH1 ERB-C (01) - PsychENCODE: Noncoding Functional Elements in the Human Brain and Their Role in the Development of Psychiatric Disorders

2018–2019	University of Michigan internal review, Searle Scholars Program
2015	UK Medical Research Council Methodology Research Panel (ad hoc)
2015	UK Biotechnology and Biological Sciences Research Council (ad hoc)
2015	Michigan Institute for Clinical and Health Research Postdoctoral Translational Scholars Program (ad hoc)

Manuscript Review

2009–present	*Ad hoc* reviewer with more than 100 verified reviews for journals including *Science*, *Nature Biotechnology*, *Nature Genetics*, *Genome Research*, *Genome Biology*, *Nature Communications*, *Bioinformatics*, *Nucleic Acids Research*, and *PLOS Computational Biology*
2012	Distinguished contributor as a leading reviewer for *Bioinformatics*

Professional Memberships

2018–present	American Society of Human Genetics
2013–present	International Society for Computational Biology
2012–present	American Association for the Advancement of Science
2005–present	Gamma Sigma Delta Agricultural Honor Society

Teaching and Mentorship

Training Programs

2024–present	**Member,** Systems & Integrative Biology Training Grant (SIB)
2021–present	**Member,** Biomedical Informatics and Data Science Training Program (BIDS-TP)
2017–present	**Member,** Cellular and Molecular Biology Program
2015–present	**Member,** Genome Science Training Program (GSTP)
2015–present	**Member,** Michigan Predoctoral Training Program in Genetics (GTP)
2014–present	**Member,** Program in Biomedical Sciences
2014–present	**Member,** Bioinformatics Training Program

Teaching (F = fall, W = winter, S = summer)

W19, W20, W21, W22, W23, W24, W25, W26	Bioinformatics Concepts and Algorithms (BIOINF 529) [Course Director]
F15, F16, F17, F18, F19, F20, F21	Gene Structure and Regulation (HUMGEN 541) [3 lectures + 2 discussions / yr.]
F19, F22	Research Responsibility and Ethics (PIBS 503) [1 discussion / yr.]
F21, W22, F24, W25	Genetics Student Seminar (HUMGEN 821/822) [Mentor]
F17, F18	Experimental Genetics Systems (HUMGEN 632) [Course Director]
F15, W16, F16, W17, F17, W18, F18	Bioinformatics Journal Club (BIOINF 602/603) [Course Director F18]
S17, S18	Introduction to Biocomputing Bootcamp (BIOSTAT/BIOINF/HUMGEN 606) [2 full days / yr.]
F15, F16, F17	Introduction to Bioinformatics & Computational Biology (BIOINF 527) [2 lectures + 3 labs / yr.]
S15, S16, S17	Basic Biology for Graduate Students with Quantitative Training (BIOINF 523) [2 lectures / yr.]
F03	Lab TA for Isotopes Tech I (Mississippi State University, BCH 4414)

Guest Lectures / Panels

2018–2019	Lecturer, REU Site: Mathematical and Theoretical Biology Institute (MTBI), Arizona State University (NSF1757968) [2 days]
2017	Panel member, University of Michigan "New Faculty Orientation to Corporate & Foundation Relations" [70 attendees]
2016	Experimental Genetics Systems (HUMGEN 632) [1 discussion]
2014	Panel member, BIOINF 527 "Challenges in Biology, Biomedicine, Data & Analysis"
2010	Co-taught Cold Spring Harbor Systems Biology Pre-meeting Workshop
2009	Duke student panelist for "How to prepare for and get into graduate school"

2008 | Taught Duke mini-course on Genome Browsers & Databases

Mentorship

Ph.D. Students (n=21)

2026–present | Kobe Howcroft (Ph.D. Student, Bioinformatics, University of Michigan)

2024–present | Ingrid Flaspohler (Ph.D. Student, Bioinformatics, University of Michigan)
NIH Biomedical Informatics and Data Science Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2024–present | Steve Losh (Ph.D. Student, Bioinformatics, University of Michigan)

2024–present | Sowmya Srinivasan (Ph.D. Student, Genetics and Genomics, University of Michigan)

2022–present | Katarina Pavlovic (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2022–present | Rintsen Sherpa (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (candidate)

2021–2025 | Kinsey Van Deynze (Ph.D. Student, Bioinformatics, University of Michigan)
NIH Genome Science Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2020–2025 | Andrea Valenzuela (Ph.D. Student, Chemical Biology, University of Michigan)
NIH Cellular Biotechnology Training Program (T32)

2020–2026 | Breanna McBean (Ph.D. Student, Genetics and Genomics, University of Michigan)
Joint M.S. in Bioinformatics, University of Michigan
NIH Genome Science Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2020–2025 | Camille Mumm (Ph.D. Student, Genetics and Genomics, University of Michigan)
Joint M.S. in Bioinformatics, University of Michigan
NIH Genome Science Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)
Rackham Predoctoral Fellowship

2018–2024 | Bradley Crone (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (candidate)

2017–2023 | Melissa Englund (Ph.D. Student, Genetics and Genomics, University of Michigan)
NIH Human Genetics Training Program (T32)
Rackham Graduate Student Research Grant (candidate)

2018–2023 | Nanxiang (Samuel) Zhao (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2016–2018 | Haley Amemiya (Ph.D. Student, Cellular and Molecular Biology, University of Michigan)
Joint M.S. in Bioinformatics, University of Michigan
NIH Cellular & Molecular Biology Training Program (T32)
NIH Cellular Biotechnology Training Program (T32) (Declined)
PIBS Excellence in Service Award
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)
Maas Professional Development Award

Rackham Graduate School Scholar-Activist Award

2016–2020 Shriya Sethuraman (Ph.D. Student, Bioinformatics, University of Michigan)

2016–2023 Christopher Castro (Ph.D. Student, Bioinformatics, University of Michigan)
NIH Bioinformatics Training Program (T32)
Rackham Merit Fellow
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)
Global Research Engagement Opportunity Fellowship

2017–2022 Ningxin Ouyang (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (candidate)

2016–2021 Shengcheng Dong (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (candidate)

2015–2021 Torrin McDonald (Ph.D. Student, Genetics and Genomics, University of Michigan)
NIH Human Genetics Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2015–2017 Greg Farnum (Ph.D. Student, Cellular and Molecular Biology, University of Michigan)

2015–2020 Sierra Nishizaki (Ph.D. Student, Genetics and Genomics, University of Michigan)
Joint M.S. in Bioinformatics, University of Michigan
NIH Genome Science Training Program (T32)
Rackham Merit Fellow
Rackham Summer Award
Rackham Graduate Student Research Grant (candidate)

M.S. Students (n=7)

2025 Christine Geng (M.S. Student, Bioinformatics, University of Michigan)
2023 Hawra Aljawad (M.S. Student, Chemical Engineering, University of Michigan)
2023 Xinyi Liu (M.S. Student, Bioinformatics, University of Michigan)
2022–2023 Emily Pogson (M.S. Student, Genetics and Genomics, University of Michigan)
2019–2020 Monica Holmes (M.S. Student, Bioinformatics, University of Michigan)
2017–2018 Nanxiang (Samuel) Zhao (M.S. Student, Bioinformatics, University of Michigan)
2015–2017 Ningxin Ouyang (M.S. Student, Bioinformatics, University of Michigan)

Additional Graduate Rotation Students (n=17)

2024 Karan Smith (Rotation Student, Cellular and Molecular Biology, University of Michigan)
2024 Jun Sik Yun (Rotation Student, Genetics and Genomics, University of Michigan)
2024 Jeremy Chen (Rotation Student, Bioinformatics, University of Michigan)
2023 Rosina Carr (Rotation Student, Bioinformatics, University of Michigan)
2023 Connor Ward (Rotation Student, Medical Scientist Training Program, University of Michigan)
2022 Brandt Bessell (Rotation Student, Bioinformatics, University of Michigan)
2022 Xiaomeng Du (Rotation Student, Bioinformatics, University of Michigan)
2022 Mahnoor Gondal (Rotation Student, Bioinformatics, University of Michigan)
2022 Xin Li (Rotation Student, Biological Chemistry, University of Michigan)
2022 Bohan Chen (Rotation Student, Cell and Developmental Biology, University of Michigan)
2021 Amelia Lauth (Rotation Student, Cellular and Molecular Biology, University of Michigan)
2019 Margarita Brovkina (Rotation Student, Cellular and Molecular Biology, University of Michigan)
2018 Steve Ho (Rotation Student, Human Genetics, University of Michigan)
2018 Matthew Pun (Rotation Student, Medical Scientist Training Program, University of Michigan)
2017 Amanda Moccia (Rotation Student, Human Genetics, University of Michigan)
2017 Stephen Carney (Rotation Student, Human Genetics, University of Michigan)
2016 Tingyang Li (Rotation Student, Bioinformatics, University of Michigan)

Postdoctoral Fellows (n=4)

2026–present Kinsey Van Deynze (University of Michigan)
2023–2025 Melissa Englund (University of Michigan)
2022–2025 Torrin McDonald (University of Michigan)
2021–2022 Shengcheng Dong (University of Michigan)

Non-student Lab Volunteers (n=2)

2019–2021 Greg Farnum (University of Michigan)
2018–2019 Monica Holmes (Postbac, University of Michigan)

Undergraduate and High School Students (n=27)

2026–present Pria Chandarana (Undergraduate, Neuroscience, University of Michigan)
2026–present Cecelia Crandall (Undergraduate, Cognitive Science, University of Michigan)
2026–present Boutheine Teyeb (Undergraduate, Computer Science, University of Michigan)
2026 Aanshi Agrawal (Undergraduate, Molecular, Cellular, and Developmental Biology, University of Michigan)
2025–2026 Krrish Thakker (Undergraduate, Biochemistry, University of Michigan)
2023–2024 Kateri Darr (Undergraduate, Computer Science, University of Michigan)
2023 Mason Miller (Undergraduate, Computer Science, University of Michigan)
2022–2024 Summer Ann (Undergraduate, Neuroscience, University of Michigan)
2022–2025 Kobe Howcroft (Undergraduate, Computer Science, University of Michigan)
2021–2024 Preston Parana (Undergraduate, UROP Molecular, Cellular, and Developmental Biology, University of Michigan)
UROP Blue Ribbon Award
2021–2022 Julia Tweadey (Undergraduate, LSA Honors Program, Life Science Informatics, University of Michigan)
2021 Aryn Booker (Undergraduate, UROP Molecular, Cellular, and Developmental Biology, University of Michigan)
UROP Blue Ribbon Award
2020 Marcela Alcaide Aligio (Undergraduate, SROP, Hunter College CUNY)
2019–2020 David Wang (Undergraduate, UROP Computer Science, University of Michigan)
2019–2020 Jack Lu (Undergraduate, UROP Computer Science, University of Michigan)
2019–2020 Diana Davis (Undergraduate, Neuroscience and German, University of Michigan)
2019 Sheila Rasouli (Undergraduate, Neuroscience, University of Toronto)
2019 Vibhasri Davuluri (High School, Girls Who Code Summer Intern)
2016–2019 Cody Morterud (Undergraduate, UROP Computer Science / Honors Capstone, University of Michigan)
2016–2017 Colten Williams (Undergraduate, UROP Computer Science, University of Michigan)
2016–2017 Courtney Asman (Undergraduate, Neuroscience, University of Michigan)
2014–2017 Maxwell Spadafore (Undergraduate, LS&A Honors Informatics, University of Michigan)
2013–2014 Natalie Ng (High School, Stanford Institutes of Medicine Summer Research)
2013–2014 Dana Wyman (Undergraduate, Biology, Stanford University)
2013 Justin Young (High School, Stanford Institutes of Medicine Summer Research)
2012 Melanie Connick (Undergraduate, Biology, University of New Mexico)
2012 Edward Dai (Undergraduate, Computer Science, Stanford University)

Doctoral Thesis Committees (n=53)

2026–present Connor Davison (Genetics and Genomics, University of Michigan, Committee Member)
2026–present Kobe Howcroft (Bioinformatics, University of Michigan, Chair)
2025–present Brandt Bessell (Bioinformatics, University of Michigan, Committee Member)
2024–present Weizhou Qian (Bioinformatics, University of Michigan, Committee Member)
2024–present Jinhao Wang (Bioinformatics, University of Michigan, Committee Member)
2024–present Brandon Klein (Medicinal Chemistry, University of Michigan, Committee Member)

2024–present Bohan Chen (Bioinformatics, University of Michigan, Committee Member)
2024–present Sowmya Srinivasan (Genetics and Genomics, University of Michigan, Co-Chair)
2024–present Ingrid Flaspohler (Bioinformatics, University of Michigan, Chair)
2024–present Steve Losh (Bioinformatics, University of Michigan, Chair)
2024–present Lu Lu (Bioinformatics, University of Michigan, Committee Member)
2023–present Linghua Jiang (Bioinformatics, University of Michigan, Committee Member)
2023–present Elysia Chou (Bioinformatics, University of Michigan, Committee Member)
2023–present Rebecca McAvoy (Molecular, Cellular, and Developmental Biology, University of Michigan, Committee Member)
2023–present Chinmay Raut (Bioinformatics, University of Michigan, Committee Member)
2022–present Katarina Pavlovic (Bioinformatics, University of Michigan, Chair)
2022–present Rintsen Sherpa (Bioinformatics, University of Michigan, Chair)
2022–present Emily Peirent (Neuroscience, University of Michigan, Committee Member)
2025–present Monica Holmes (Bioinformatics, University of Michigan, Committee Member)
Single-Molecule Long-read Sequencing for Resolution of Structurally Complex Genomic Loci and Accurate Phenotype Prediction
2024–2026 Matthew Hodgman (Bioinformatics, University of Michigan, Committee Member)
Raising the Standard of Machine Learning Transparency and Utility in Cardiovascular Medicine
2020–2026 Breanna McBean (Genetics and Genomics, University of Michigan, Co-Chair)
Targeting Resistance and Metastasis: Advancing Local and Systemic Therapeutic Strategies for Aggressive Breast Cancers
2021–2026 Mashiat Rabbani (Genetics and Genomics, University of Michigan, Committee Member)
A Genome-scale Program for Building the Sperm Epigenome
2024–2025 Lingrui Cai (Bioinformatics, University of Michigan, Committee Member)
Advancing Medical Image and Report Analysis with Artificial Intelligence: Applications in Inflammatory Bowel Disease and Traumatic Brain Injury
2021–2025 Kinsey Van Deynze (Bioinformatics, University of Michigan, Chair)
Sequence Analysis of Tandem Repeats Utilizing 3rd Generation Sequencing Technologies
2020–2025 Andrea Valenzuela (Chemical Biology, University of Michigan, Co-Chair)
AI-Driven Workflows Integrating Strain Diversity and Host Metabolic-Genetic Variation for Tuberculosis Treatment Prioritization
2021–2025 Zhenhao Zhang (Bioinformatics, University of Michigan, Committee Member)
Decoding Genome Regulation through Deep Learning Models
2020–2025 Steve Ho (Genetics and Genomics, University of Michigan, Committee Member)
Applied Machine Learning for Big Data Genomics
2020–2025 Camille Mumm (Genetics and Genomics, University of Michigan, Chair)
Exploring Repetitive Elements in Neurodegenerative Disease Using Targeted Long-Read Sequencing
2022–2025 Kaiwen Deng (Bioinformatics, University of Michigan, Committee Member)
Advancing Computational Models and Algorithms for the Analysis of Single-Cell and Neural Data
2022–2024 Franco Tavella (Biophysics, University of Michigan, Committee Member)
Robustness and Tunability of Biological Oscillations
2018–2024 Bradley Crone (Bioinformatics, University of Michigan, Chair)
Computational Methods in Functional Prioritization of Polygenic Risk Score Models
2021–2024 Wenjin Gu (Bioinformatics, University of Michigan, Committee Member)
Development of Viral Integration Analysis Technologies for Virus-Associated Cancer Research
2018–2023 Rucheng Diao (Bioinformatics, University of Michigan, Committee Member)
Local Chromatin Environments Shape Transcription and Adaptive Immunity in Bacteria
2021–2023 Zijun Gao (Bioinformatics, University of Michigan, Committee Member)
Advance Machine Learning and Image Analysis Methods for Clinical Decision Support in Cardiovascular and Pulmonary Diseases
2018–2023 Nanxiang (Samuel) Zhao (Bioinformatics, University of Michigan, Chair)
Decoding Regulatory Variants with Computational Methods in Non-coding Regions of the Human Genome
2020–2023 Ashley Melnick (Cellular and Molecular Biology, University of Michigan, Committee Member)
Cdc73 Protects Notch-Induced Leukemia Cells From DNA Damage and Mitochondrial Stress
2016–2023 Christopher Castro (Bioinformatics, University of Michigan, Chair)

Investigating the Role of Noncoding De Novo Single-Nucleotide Variants in Autism Spectrum Disorder

2017–2023 Melissa Englund (Genetics and Genomics, University of Michigan, Chair)
Identification and Characterization of Cis-Regulatory Elements in the Human Genome

2018–2023 Stephen Carney (Cancer Biology, University of Michigan, Committee Member)
Epigenetic reprogramming in mutant IDH1 glioma influences radioresistance and neural lineage differentiation

2019–2023 Benjamin Yang (Biomedical Engineering, University of Michigan, Committee Member)
Towards Defining Principles of Cell Fate Plasticity

2018–2022 Marcus Sherman (Bioinformatics, University of Michigan, Committee Member)
Cultivation of enhanced bioinformatic-specific pedagogical manipulatives, interventions, and professional development

2021–2022 Kuan-Han Hank Wu (Bioinformatics, University of Michigan, Committee Member)
Integrating Electronic Health Records with Genetic Information to Advance Precision Medicine Approaches in Cardiovascular Disease

2017–2022 Amanda Moccia (Genetics and Genomics, University of Michigan, Committee Member)
Investigation of Developmental Disorders: Genetic Discovery and Functional Validation

2017–2022 Ningxin Ouyang (Bioinformatics, University of Michigan, Chair)
Deciphering Transcriptional Regulatory Circuits: Transcription Factor Binding and Regulatory Variants Identification

2015–2021 Torrin McDonald (Genetics and Genomics, University of Michigan, Chair)
Leveraging New Technologies to Explore Regulatory and Structural Elements of the Human Genome

2018–2021 Heming Yao (Bioinformatics, University of Michigan, Committee Member)
Machine Learning and Image Processing for Clinical Outcome Prediction: Applications in Medical Data from Patients with Traumatic Brain Injury, Ulcerative Colitis, and Heart Failure

2016–2021 Mohd Hafiz Bin Mohd Rothi (Molecular, Cellular, and Developmental Biology, University of Michigan, Committee Member)
Control of Chromatin by RNA-mediated Transcriptional Silencing

2016–2021 Shengcheng Dong (Bioinformatics, University of Michigan, Chair)
Computational Methods to Identify Regulatory Variants in the Non-coding Regions of the Human Genome

2017–2021 Steven Romanelli (Molecular & Integrative Physiology, University of Michigan, Committee Member)
Viral CRISPR/Cas9 Gene Transfer for Somatic Knockout in Brown Adipose Tissue

2018–2021 Negar Farzaneh (Bioinformatics, University of Michigan, Committee Member)
Automated Decision Support System for Traumatic Injuries

2016–2020 Shriya Sethuraman (Bioinformatics, University of Michigan, Co-Chair)
Genome-wide Identification of Non-coding Transcription by RNA Polymerase V and Its Involvement in Transcriptional Gene Silencing

2015–2020 Sierra Nishizaki (Genetics and Genomics, University of Michigan, Chair)
Decoding the Non-coding Genome: Novel Technologies for the Characterization of Non-coding Elements and Variation

2017–2020 Christopher Lee (Biostatistics, University of Michigan, Committee Member)
Improvements and Developments in Gene Regulation and Single-Cell Gene Expression Data Analysis

2018–2019 Christine Ziegler (Biological Chemistry, University of Michigan, Committee Member)

2015–2018 Ari Allyn-Feuer (Bioinformatics, University of Michigan, Committee Member)
The Pharmacoepigenomics Informatics Pipeline and H-GREEN Hi-C Compiler: Discovering Pharmacogenomic Variants and Pathways with the Epigenome and Spatial Genome

2015–2017 Raymond Cavalcante (Bioinformatics, University of Michigan, Committee Member)
Beyond the Transcriptome: Facilitating Interpretation of Epigenomics and Metabolomics Data

2015–2017 Zhengting Zou (Bioinformatics, University of Michigan, Committee Member)
Model-based genomic studies of protein sequence evolution: convergence, epistasis, and amino acid acceptance rates

Preliminary Exam Committees (n=46)

2026 James George (Cellular and Molecular Biology, University of Michigan)
2025 Xinyi Deng (Bioinformatics, University of Michigan)
2025 Mitchell Witt (Bioinformatics, University of Michigan)
2024 Luke Gohmann (Cellular and Molecular Biology, University of Michigan)
2024 Benjamin Li (Bioinformatics, University of Michigan)
2024 Tiffany Wan (Bioinformatics, University of Michigan)
2024 Zhiyuan Yu (Bioinformatics, University of Michigan)
2024 Rebecca McAvoy (Molecular, Cellular, and Developmental Biology, University of Michigan)
2024 Bonje Obua (Cellular and Molecular Biology, University of Michigan)
2024 Abigail Vallie (Cellular and Molecular Biology, University of Michigan)
2023 Jinhao Wang (Bioinformatics, University of Michigan)
2023 Lishi Yin (Bioinformatics, University of Michigan)
2023 Matthew Hodgman (Bioinformatics, University of Michigan)
2023 Ilakkiya Venkatachalam (Genetics and Genomics, University of Michigan)
2023 Jianhui Gong (Bioinformatics, University of Michigan)
2023 Mahnoor Gondal (Bioinformatics, University of Michigan)
2023 Elysia Chou (Bioinformatics, University of Michigan)
2022 Sean Moran (Bioinformatics, University of Michigan)
2022 Lu Lu (Bioinformatics, University of Michigan)
2022 Linghua Jiang (Bioinformatics, University of Michigan)
2022 Kaiwen Deng (Bioinformatics, University of Michigan)
2022 Yufeng Zhang (Bioinformatics, University of Michigan)
2021 Anthony Nguyen (Human Genetics, University of Michigan)
2021 Hanbyul Cho (Bioinformatics, University of Michigan)
2021 Charles Ryan (Cellular and Molecular Biology, University of Michigan)
2021 Kuan-Han Wu (Bioinformatics, University of Michigan)
2021 Wenjin Gu (Bioinformatics, University of Michigan)
2020 Jie Cao (Bioinformatics, University of Michigan)
2020 Zijun Gao (Bioinformatics, University of Michigan)
2020 Ashley Melnick (Cellular and Molecular Biology, University of Michigan)
2019 Benjamin Yang (Biomedical Engineering, University of Michigan)
2019 Maria Virgilio (Cellular and Molecular Biology, University of Michigan)
2018 Zhi Carrie Li (Bioinformatics, University of Michigan)
2018 Kevin Hu (Bioinformatics, University of Michigan)
2018 Siyu Liu (Bioinformatics, University of Michigan)
2018 Alexandra Weber (Bioinformatics, University of Michigan)
2018 Mitch Fernandez (Bioinformatics, University of Michigan)
2017 Tingyang Li (Bioinformatics, University of Michigan)
2017 Marcus Sherman (Bioinformatics, University of Michigan)
2017 Adrienne Shami (Human Genetics, University of Michigan)
2017 Trenton Frisbie (Human Genetics, University of Michigan)
2017 Melissa Englund (Human Genetics, University of Michigan)
2017 Peter Orchard (Bioinformatics, University of Michigan)
2017 Li Guan (Bioinformatics, University of Michigan)
2016 Shriya Sethuraman (Bioinformatics, University of Michigan)
2016 Jed Carlson (Bioinformatics, University of Michigan)

Publications

* co-first authorship; † co-senior authorship; underline indicates Boyle Lab members

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